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Impact of micro and macroporous TFF membranes on product sieving and chromatography loading for perfusion cell culture

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Abstract

Bioprocess intensification can be achieved through high cell density perfusion cell culture with continuous protein capture integration. Protein passage and cell retention are commonly accomplished using tangential flow filtration systems consisting of microporous membranes. Significant challenges, including low efficiency and decaying product sieving over time, are commonly observed in these cell retention devices. Here, we demonstrate that a macroporous membrane overcomes the product sieving challenges when comparing to several other membrane chemistries and pore sizes within the microporous range. This way, variable chromatography column loading is avoided. The macroporous membrane yielded a 13,000 L/m² volumetric throughput. The membrane's cut-off size results in an increased permeate turbidity due to particles passage, such as cell debris, through pores ranging from 1 to 4 µm. In addition, successful chromatography column plugging mitigation was achieved by employing depth filtration before the chromatographic step. Depth filtration volumetric throughputs were between 600 and 1,000 L/m². Combining a macroporous cell retention device with a depth filter not only provided an alternative to address the challenge of undesired long protein residence times in the bioreactor due to product sieving decay, but also exhibited a throughput increase, making the integration of multicolumn capture chromatography with a perfusion cell culture a more robust process.

KEYWORDS

Chinese hamster ovary, integrated continuous bioprocessing, monoclonal antibody, product sieving, tangential flow filtration perfusion

1 | INTRODUCTION

The biopharmaceutical landscape with diverse portfolios and varied production needs drives innovation towards more cost effective and flexible manufacturing processes. Next generation technologies based on perfusion cell culture are capturing manufacturers' attention with higher volumetric productivities than traditional production modes (Klutze et al., 2015; Rathore et al., 2015; Yang et al., 2016). Compared to fed-batch, perfusion culture by its very nature uses substantially larger volumes of cell culture media which must generally be clarified before the downstream capture step.

Therefore, bioprocess intensification via integration of perfusion cell cultures with continuous multicolumn capture chromatography is desired. Continuous capture of therapeutic proteins from a perfusion cell culture with a continuous purification process not only eliminates large hold tanks, but also enhances resin utilization (Somasundaram et al., 2018; Warikoo et al., 2012).

In filtration-based perfusion cell culture, membrane fouling and product sieving are the major drawbacks (Belfort et al., 1994; Bolton & Apostolidis, 2017; M.-m. Kim & Zydne, 2006; Zydne, 2016). These can lead to diminished permeate flow forcing an abrupt batch termination, or increased residence time of the therapeutic proteins

in the bioreactor (van Reis & Zydney, 2007). The latter may detrimentally affect the target product quality attributes, as well as product recovery. One strategy to overcome these challenges seeks to avoid the release of membrane foulants originating from cell lysis in the first place (Mariorella et al., 1991; Vickroy et al., 2007). Xu et al. (2017) showed that increases in the shear protectant concentration for 200-L perfusion cell cultures lead to higher cell viability and improved product sieving. Other researchers' have studied the mechanical stress caused by pumps for cell culture broth recirculation through the cell retention device (Blaschczok et al., 2013; Karst et al., 2016; Wang et al., 2017). These studies suggest that cellular damage was more severe using a peristaltic pump than lower hydrodynamic shear versions, such as diaphragm or centrifugal magnetically-levitated pumps. Several of the previous tangential flow filtration (TFF) based perfusion literature studies (Brennan et al., 1987; Clincke, Mölleryd, Samani, et al., 2013; Clincke, Mölleryd, Zhang et al., 2013; de la Broise et al., 1992; de la Broise, Noiseux, Lemieux & Massie, 1991; Zhang et al., 1993) employed high shear peristaltic pumps for cell culture fluid recirculation which may result in worst case scenario filtration performances. Nonetheless, there are limited literature references where low shear recirculating pumps were couple to TFF units for mammalian cell culture. In addition, a technical gap remains in addressing product sieving causing protein retention in these unidirectional TFF perfusion systems, as well as the integration of perfusion with continuous capture chromatography.

As suggested by Wang et al. (2017), particles in the perfusion cell culture broth of approximately 0.1 μm size (within 100 nm order of magnitude range) are responsible for the presence of product sieving in microporous TFF membranes. Here, we propose to use macroporous membranes to allow passage of these particles, and this way avoid product sieving in TFF perfusion. This approach should not hinder the direct integration with downstream processes, but it does require additional filtration to remove the particles from the perfusion permeate stream to avoid column plugging.

This work investigates several TFF perfusion membrane chemistries and pore sizes within the microporous range (0.2 and 0.5 μm), as well as a macroporous membrane with a cut-off range between 1 and 4 μm . To accomplish this, a single mammalian perfusion cell culture bioreactor was connected to multiple TFF membranes to assess the different characteristics. Here, we demonstrate that the macroporous TFF membrane overcomes the product sieving challenges on a head-to-head comparison with microfiltration membranes exposed to the same cell culture material. This finding is then confirmed with two additional experiments using a biomanufacturing relevant TFF perfusion system. To investigate the lifetime of the macroporous perfusion membrane, one experiment was conducted at 40 $\text{L}\cdot\text{m}^{-2}\cdot\text{hr}^{-1}$ permeate flux with 80% recycle, while maintaining the desired bioreactor exchange rate. The volumetric throughput obtained was 13,000 L/m^2 after 20 culture days. Finally, To address the particle passage through the macroporous perfusion membrane, depth filtration was incorporated in a 45-day perfusion cell culture integrated with multi-column protein A chromatography. Initially, the

cell retention device used a microfiltration membrane. On the 28th day of cell culture, the perfusion membrane was replaced by a macrofiltration membrane allowing for the depth filters evaluation which yielded volumetric throughputs between 600 and 1000 L/m^2 . This is the first time, to our knowledge, that a TFF-based perfusion system demonstrated high product sieving (i.e., no product retention) for a continuous antibody production process integrated with capture chromatography.

2 | MATERIALS AND METHODS

2.1 | Perfusion cell culture

In this work, all cell cultures produced a proprietary recombinant monoclonal antibody expressed in a CHO cell line using commercially available, chemically-defined media. These cultures were supplemented with glucose (maintained ≥ 2 g/L) and antifoam as needed. pH was controlled at 7.10 ± 0.35 and the dissolved oxygen was maintained at 30% of air saturation with 30 W/m^3 constant agitation. The target inoculation for all cultures was 0.5×10^6 cells/ml. Cell culture metabolites, as well as lactate dehydrogenase (LDH), were determined using a YSI Model 2700 analyzer (Yellow Springs Instruments, OH) and Randox Rx Daytona (Randox laboratories, United Kingdom). Trypan blue dye exclusion method was used to measure viable cell density (VCD) offline (Cedex cell counter, Roche, Mannheim, Germany) and capacitance to determine VCD online (Incyte, Hamilton, Reno, NV). Turbidity was measured using a Hach 2100N turbidimeter (Hach, Loveland, CO).

2.2 | TFF membrane screening

Figure 1a shows a schematic of the experimental setup for the membrane screening studies. A cell culture bioreactor (Sartorius AG, Goettingen, Germany) was connected to five TFF membranes (see Table 1). Medium exchange was started on the 2nd day of cell culture. Each TFF membrane was operated at constant permeate flux and 1500 s^{-1} shear rate. A TFF system (PendoTECH, Princeton, NJ) gravimetrically controlled the bioreactor at a constant level using a Explorer Pro scale (Ohaus, Parsippany, NJ) by feeding cell culture media with a Masterflex pump (Cole-Parmer, Vernon Hills, IL).

The TFF membrane modules were attached to the bioreactor through kleenpak sterile connectors (Pall, Port Washington, NY) on an autoclaved manifold made in-house with Masterflex silicone L/S 15 and PharMed BPT L/S 16 tubing (Cole-Parmer, Vernon Hills, IL). Each of the independently operated TFF membranes required one Masterflex pump (Cole-Parmer, Vernon Hills, IL) for cell culture broth recirculation and one ChemTec CP-8 pump (SciLog, Madison, WI) for the permeate withdrawal. In contrast to the other perfusion experiments in this work, a cell discard rate was absent for this membrane screening. The focus was to create a more challenging perfusion scenario by removing the cell bleed and including peristaltic pumps for the culture recirculation (i.e., lower viability cultures).

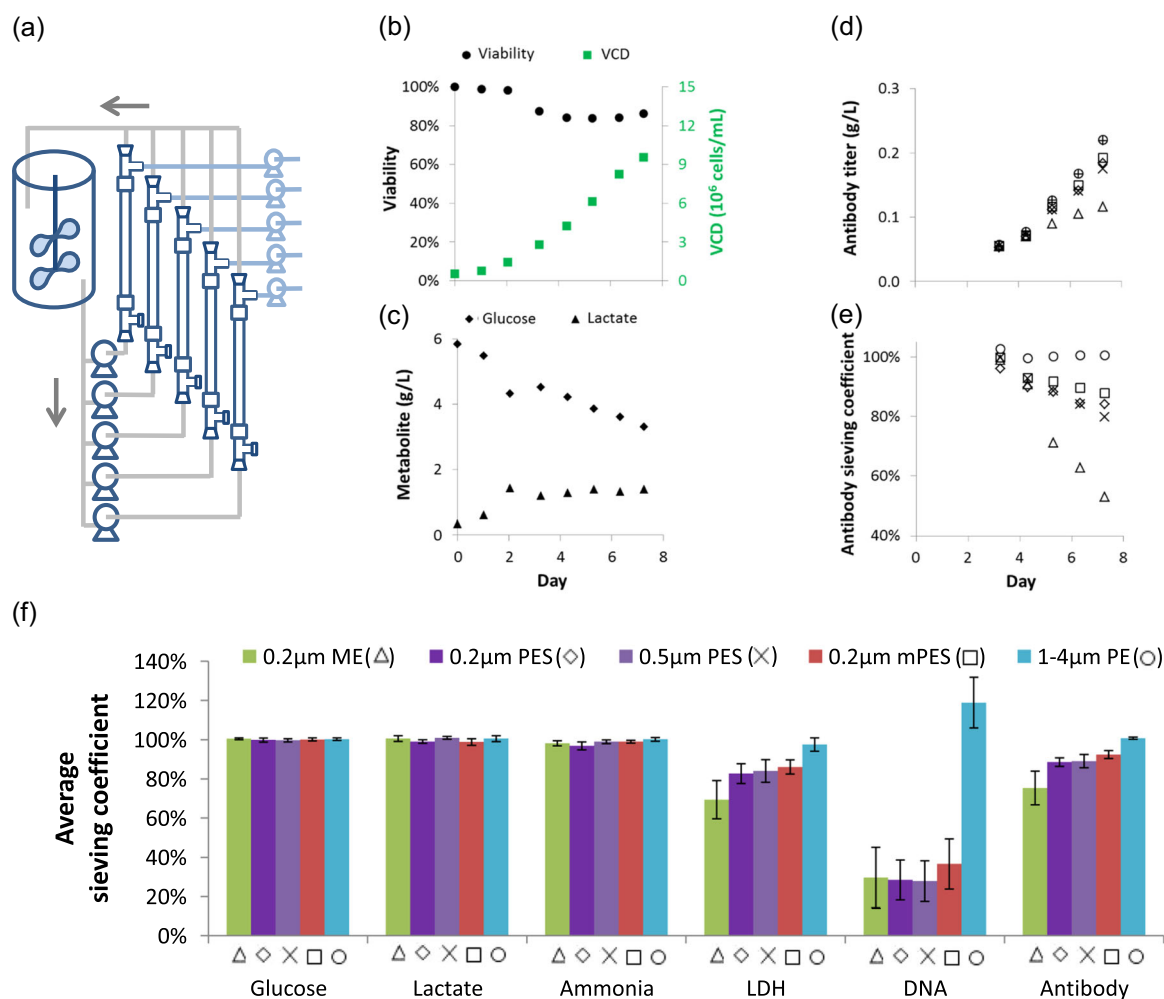


FIGURE 1 Perfusion membrane screening with one bioreactor connected to five perfusion membranes: (a) schematic illustration of the membranes and manifold; (b) viability and viable cell density (VCD) of the perfusion cell culture; (c) glucose and lactate concentration in the bioreactor; (d) antibody titer for bioreactor (+) and permeate from all membranes; (e) daily antibody sieving coefficient per membrane; (f) average sieving coefficient for metabolites and biomolecules for each membrane (membrane details in Table 1) [Color figure can be viewed at wileyonlinelibrary.com]

2.3 | Macroporous TFF membrane evaluation

A KML-100 TFF perfusion system with a low shear magnetically-levitated pump and a macroporous TFF filter both from Spectrum Laboratories (Repligen, Waltham, MA) were employed. A capacitance-based bleed was used to maintain an operating cell density of $75 \pm 10 \times 10^6$ cells/mL. The perfusion loop operated at 1500 s^{-1} shear rate and total permeate flux of $40 \text{ L} \cdot \text{m}^{-2} \cdot \text{hr}^{-1}$. Medium exchange started on the 3rd day of cultivation at 0.75 VVD perfusion rate with an additional 80% permeate recycle. The experiment duration to evaluate the macroporous membrane's volumetric throughput was reduced from several months to 3 weeks by recycling the permeate stream.

2.4 | TFF-based perfusion integrated with capture chromatography

A KML-100 perfusion system was employed. Medium exchange was started on the 3rd day of cultivation at 0.75 VVD targeting an

operating cell density of $75 \pm 10 \times 10^6$ cells/mL with a capacitance-based cell bleed. On the 28th culture day, the microporous perfusion membrane was exchanged to a macroporous membrane using a BioWelder Total Containment (Sartorius AG, Goettingen, Germany). The depth filters employed were Emphaze AEX Hybrid Purifier (3 M, Meriden, CT), Supracap 100 PDE2 (Pall, Port Washington, NY) and Millistak A1HC (Merck KGaA, Darmstadt, Germany) with part numbers EMP513AEX020A, NP5LPDE21 and MA1HC027H1, respectively. The downstream integration with the multi-column chromatography system Cadance BioSMB (Pall, Port Washington, NY) was previously described by Patel et al. (2017).

2.5 | Analytical methods

A gradient elution chromatography method was used to determine antibody concentrations. The chromatographic column employed was 2.1 mm I.D. $\times$ 30 mm length POROS R2/10 reverse-phase column (Life Technologies, Carlsbad, CA). Liquid phase reagents (Thermo Fisher Scientific, Fair Lawn, NJ) used were acetonitrile, water and

TABLE 1 Membrane characteristics and process conditions for all perfusion cell culture experiments

Experiment	Perfusion membrane characteristics			Perfusion process conditions		
	Part number	Chemistry	Area (cm ²)	Pore size (μm)	Permeate flux (L·m ⁻² ·hr ⁻¹)	Media exchange (VVD)
Perfusion membrane screening	D02-P20U-10-N	PES	88	0.2	2	1
	D02-E20U-10-N	mPES	75	0.2		
	D02-M20U-10-N	ME	94	0.2		
	C02-P50U-10-N	PES	13	0.5		
	P-D04-R02M-03-N	PE	39	1-4		
Macroporous membrane evaluation	P-N04-R01M-03-8L	PE	330	1-4	40	0.75
Perfusion & continuous chromatography integration	N04-E20U-10	mPES	3300	0.2	0.8	0.75
	P-N04-R01M-03-8L	PE	330	1-4	8	

Abbreviations: ME, mixed cellulose ester; mPES, modified polyethersulfone; PES, polyethylene; PE, polyethersulfone; VVD, vessel volume per day (day⁻¹).

trifluoroacetic acid. High molecular weight species (HMW) were determined by size exclusion chromatography (SEC) with a 4.6 mm I.D. × 150 mm length Acquity UPLC BEH200 column (Waters, Milford, MA). SEC liquid phase reagents were arginine and sodium phosphate (Life Technologies, Carlsbad, CA). DNA was determined by the quantitative polymerase chain reaction method developed at Merck & Co., Inc. (Kenilworth, NJ).

2.6 | Computational equations

The perfusion cell culture medium exchange rate (D) and volumetric productivity (VP) are defined, as suggested by Konstantinov et al. (2006), based on the permeate flow rate (Q_{Perm}):

$$D = Q_{\text{Perm}} / V_{\text{BRX}} \quad (1)$$

$$VP = D \times C_{\text{Perm}} \quad (2)$$

where D and Q_{Perm} are expressed in vessel volumes per day (VVD) and L/day, respectively. The bioreactor's working volume (V_{BRX}) and permeate concentration (C_{Perm}) are expressed in L and g/L, respectively. For the TFF filtration performance assessment, the sieving coefficient for any species is defined as follows:

$$\text{Sieving coefficient (\%)} = C_{\text{Perm}} / C_{\text{BRX}} \times 100 \quad (3)$$

where C_{BRX} is the species bioreactor concentration in g/L. The transmembrane pressure (TMP) used to evaluate the volumetric throughput of the perfusion membrane is defined as follows (van Reis et al., 1991):

$$\text{TMP} = \left(\frac{P_F + P_R}{2} \right) - P_P \quad (4)$$

where the feed, retentate, and permeate pressures for the perfusion membrane are P_F , P_R , and P_P , respectively. The equation employed to calculate the wall shear rate (γ) of the perfusion membrane lumens is defined as follows:

$$\gamma = \frac{8v_{\text{CF}}}{d_{\text{HF}}}, \quad (5)$$

where v_{CF} is the hollow fiber's crossflow velocity and d_{HF} is the hollow fiber's lumen diameter.

3 | RESULTS

3.1 | TFF membrane screening

A perfusion bioreactor connected to a set of five TFF membranes, shown in Figure 1a, was run for 7 days. The (b) and (c) panels of Figure 1 show viable cell density and metabolite concentration profiles of healthy cell cultures in exponential growth phase. The impact of using a manifold and peristaltic pumps for cell culture broth recirculation is observed on Figure 1b, where an initial viability decay is observed on the Day 3 corresponding to the media exchange start.

This observation is in-line with the Blaschczok et al. (2013) study on the impact of peristaltic pumps on cell culture viability. The increased mechanical stress by the peristaltic pump leads to the release of intracellular components and an increase of cell debris creating a more challenging filtration scenario. This way, the mechanism of membrane fouling is accelerated and antibody sieving coefficient differences can be assessed within a week (Figure 1d,e).

The average sieving coefficient for antibody, LDH, DNA, and several metabolites are shown in Figure 1f. These factors were monitored to investigate the potential impact of different molecule characteristics on the performance of the perfusion membranes under study. Variations in low molecular weight metabolite concentrations not only allow an assessment of the potential impact of the species size, but also can affect cell culture behavior when byproducts accumulate. Within the normal range of pH's encountered in cell culture (7.1 ± 0.35), LDH has approximately the same size as the antibody studied (molecular weight ≈ 150 kDa), but the opposite charge (negative and positive, respectively). These factors enabled the evaluation of protein charge as a variable in antibody sieving decay. Lastly, DNA concentrations were measured as the DNA sieving coefficient decay has been suggested to be associated with antibody sieving coefficient decay (Mercille et al., 1994).

A TFF-based perfusion cell culture was used to study the impact of several membrane chemistries and pore sizes, including micro and macroporous membranes, on antibody sieving. As shown in Figure 1e, 0.2- μm pore size membrane with mixed cellulose ester chemistry (ME) showed the sharper antibody sieving coefficient decay, suggesting that cellulose-based membrane chemistries underperform in perfusion cell culture. Whereas the family of polyethersulfone (PES and modified PES-mPES) membranes with pore sizes between 0.2 and 0.5 μm clustered together with an intermediate antibody sieving coefficient slope. The macroporous perfusion membrane was the only membrane studied that showed no sieving coefficient decay for the antibody product, LDH and DNA (Figure 1f). Both antibody and LDH, with similar size, showed consistent average sieving coefficient decay across all membranes tested suggesting that charge is not playing a significant role in membrane performance. The cellular metabolites, approximately 1000-fold smaller than the DNA and antibody, showed high sieving coefficient on all the membranes tested regardless of ionic charge. The average DNA sieving coefficient profile show an accentuated decay for all membranes except the macroporous membrane. Moreover, DNA may be used as a fouling marker and it is possible that DNA may be involved in the fouling mechanism as suggested by Mercille et al. (1994).

3.2 | Evaluation of macroporous perfusion membrane's volumetric throughput

To mimic a large scale manufacturing relevant scenario, a low shear, single-use TFF perfusion system was used to assess the macroporous TFF perfusion membrane's lifetime. Figure 2 shows, under the near steady-state conditions over the last 10 days of cultivation, the averaged cell density was $68.5 \pm 6.5 \times 10^6$ cells/ml with $98.8 \pm 0.8\%$

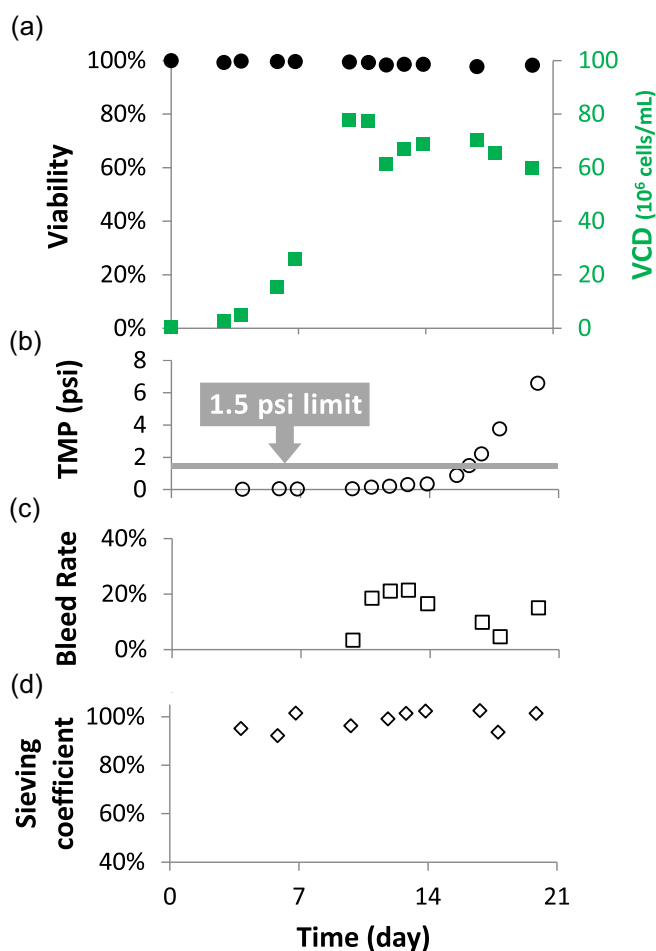


FIGURE 2 Macroporous perfusion membrane lifetime evaluated till 1.5 psi transmembrane pressure: (a) viability and viable cell density of the perfusion cell culture; (b) perfusion membrane transmembrane pressure; (c) cell culture bleed rate in percent of bioreactor working volume per day; (d) antibody sieving coefficient [Color figure can be viewed at wileyonlinelibrary.com]

cell culture viability (Figure 2a). A transmembrane pressure (TMP) of 1.5 psi was used to determine the membrane's volumetric throughput of 13,000 L/m² (Figure 2b). This volumetric throughput is significantly higher than the 1000 and 5000 L/m² that Clincke, Mölleryd, Zhang et al. (2013) reported for microporous TFF-based perfusion. Although arbitrary, the 1.5 psi cut-off used in this comparison would allow for a perfusion filter exchange within a 2 day window from the time of the warning. Lastly, these process conditions are a worst case scenario with regard to the permeate flux ($40 \text{ L} \cdot \text{m}^{-2} \cdot \text{hr}^{-1}$), a higher volumetric throughput may be obtained at lower permeate fluxes.

3.3 | Effect of membrane characteristics on the integrated capture chromatography step

To evaluate the impact of perfusion membranes on integrated continuous bioprocessing, a bioreactor was connected to capture chromatography while employing different perfusion membranes. Initially a microporous perfusion membrane was employed. On Day

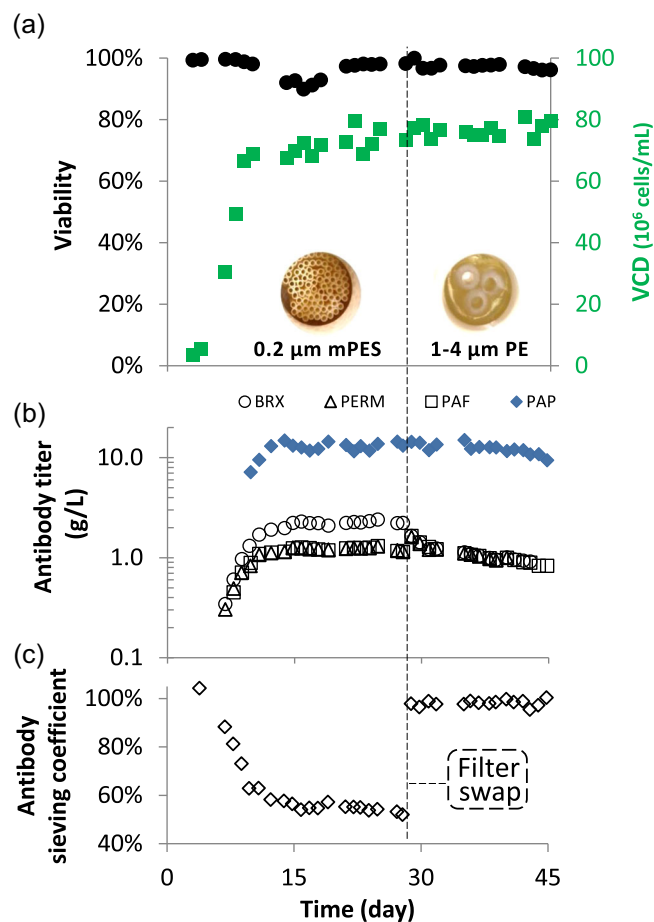


FIGURE 3 Perfusion cell culture integrated with capture chromatography, initially performed with a microporous membrane (0.2 μm) and swapped on the 28th culture day to a macroporous perfusion membrane (1–4 μm), as demonstrated by the dashed line: (a) perfusion cell culture viability and vessel volume per day; (b) antibody titer in the bioreactor (BRX), permeate (PERM), protein A capture chromatography feed (PAF), and protein A product (PAP); (c) antibody sieving coefficient [Color figure can be viewed at wileyonlinelibrary.com]

28th of cultivation, the perfusion membrane was replaced with a macroporous membrane. The mid-run membrane swap allowed direct assessment of the impact of using microporous versus macroporous membranes on (a) perfusion cell culture performance, (b) depth filtration needs before the chromatography step, and (c) antibody sieving coefficient as shown in Figures 3–5, respectively.

The average cell density was controlled at $73.6 \pm 3.5 \times 10^6$ cells/ml with an average cell culture viability of $96.4 \pm 2.8\%$ over the last 30 days of cultivation (Figure 3a). The antibody concentration in the bioreactor was significantly higher than permeate concentration whereas employing a microporous membrane (Figure 3b). In contrast, when employing the macroporous membrane (Figure 3c), similar protein concentration were observed for both the bioreactor and permeate samples and the sieving coefficient remained high (without showing a decay). This observation of high product sieving was observed despite the fact that the macroporous membrane had a 10-fold smaller nominal surface area. Figure 3b shows the concentration

of the protein A product (PAP, i.e., protein A elution) demonstrating consistency over time in the product capture integrated with perfusion cell culture.

Due to the 1–4 μm pore size range, the macroporous perfusion membrane allows particles within the 100 μm range and larger colloids to permeate. Unlike the macroporous membrane, microporous membranes can retain these particles within bioreactors, which have been identified as potential cause for product sieving coefficient decay and lead to fouling both in perfusion cell culture (Wang et al., 2017) as well as in primary recovery (Roush & Lu, 2008). When employing the microfiltration perfusion membrane, no extra filtration is required between the bioreactor and the capture chromatography step, as shown by the low turbidity on the early stage of cell culture (Figure 4b). However, when employing the macroporous membrane, the permeate turbidity measured varied between 65 and 80 NTU (Figure 4b after Day 28). Levels above 10 NTU lead to chromatography column clogging and require an additional filtration. Figure 4 shows relevant data regarding the depth filtration loadings achieved with the macroporous membrane (Figure 4a) and a schematic of the integrated perfusion and capture chromatography process (images of the surge vessels between unit operations were taken whereas employing a macroporous perfusion membrane to highlight the decreased turbidity across the depth filtration).

The criteria for maximum volumetric throughput of the depth filtration at feed pressure of 15 psi or maximum turbidity of 10 NTU was evaluated for different depth filters. The volumetric throughputs obtained were 757, 970, and 643 L/m^2 for Emphaze, PDE2, and A1HC depth filters, respectively (Figure 4a). After Day 28, the reduction in permeate turbidity was accomplished while evaluating the depth filtration volumetric throughput as represented by the arrow shown in Figure 4b. Employing a 0.2- μm dead-end filter was not considered since initial studies suggested that the loadings were too low for process feasibility ($<100 \text{ L}/\text{m}^2$ —data not shown). Requiring an extra filtration step is a disadvantage of using macroporous perfusion membranes, not only due to the extra cost added but also due to the higher sterility risk. Integrated processes require an aseptic envelope between the bioreactor and the capture step to avoid bioburden ingress. The growth promoting spent cell culture media constantly going through the depth filtration require these filters to be sterile to minimize bioburden risk. Depth filtration is traditionally used in batch processing with an operating window of hours, whereas continuous integrated bioprocesses operate on a time scale of days. Continuous integrated bioprocesses therefore need more options for sterile depth filtration with higher loading to also minimize the frequency of changeouts.

The impact of employing the micro and macroporous perfusion membranes on sieving coefficient for antibody and DNA shown in Figure 5a is in agreement with the TFF membrane screening studies (Figure 1f), where the decay is only observed with the microporous membrane. Moreover, no statistically significant difference was observed on the product quality attributes or purification process yield of material produced by the two perfusion membranes even with a decaying sieving coefficient (Figure 5b). With a constant bleed

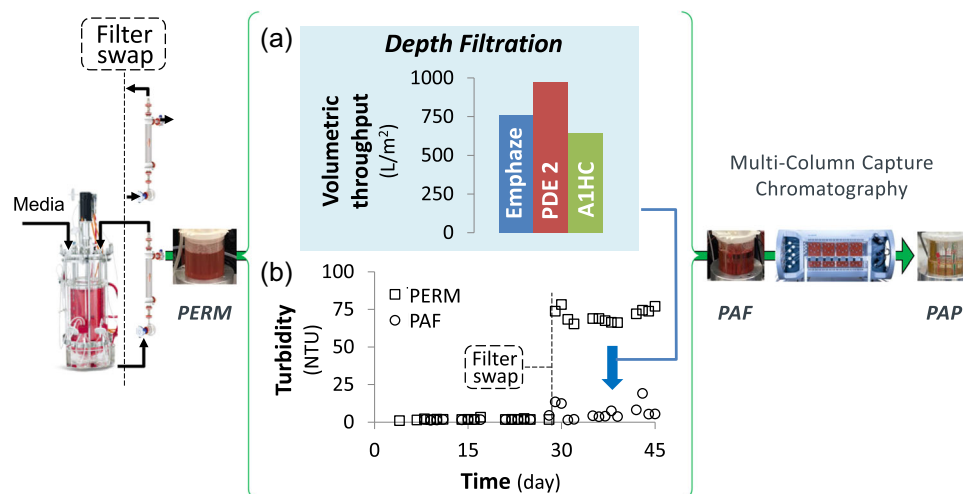


FIGURE 4 Integration of the perfusion cell culture with continuous capture chromatography (protein A): (a) Depth filtration volumetric throughput till feed pressure of 15 psi or PAF turbidity of 10 NTU; (b) Turbidity for PERM and PAF surge vessels. Images of the actual surge vessels between unit operations are shown to highlight the turbidity. Process locations are permeate (PERM), protein A feed (PAF) and protein A product (PAP) [Color figure can be viewed at wileyonlinelibrary.com]

rate maintaining the target operating cell density, the perfusion volumetric productivity was comparable for both perfusion membranes. As evidenced by no increase in high molecular weight species, increased bioreactor residence time due to the decaying antibody

sieving coefficient did not lead to a significant difference on the antibody critical quality attributes. Nonetheless, even though no product quality impact was observed for the antibody studied, other more liable molecules may be impacted (S. C. Kim et al., 2016).

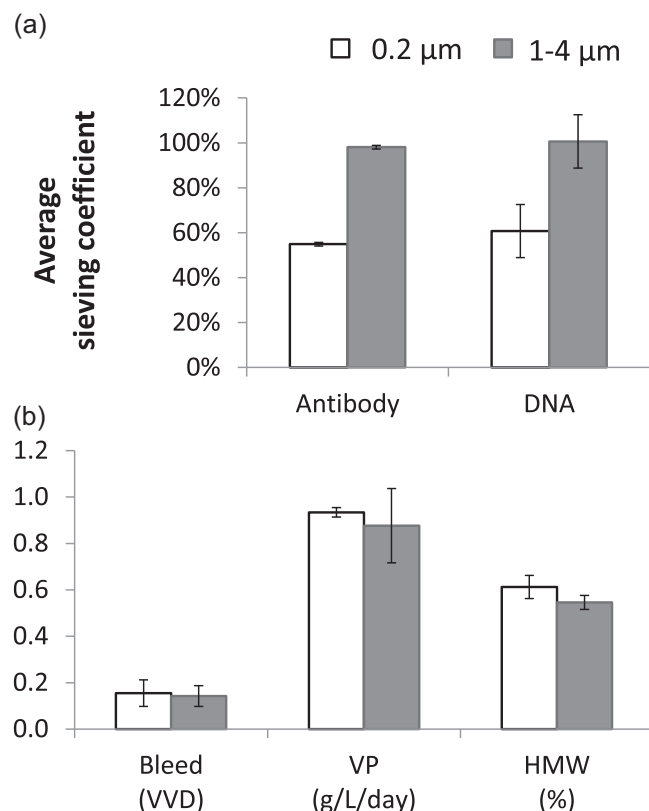


FIGURE 5 (a) Average antibody (perfusion cell culture also shown in Figure 3) and DNA sieving coefficient for microporous (0.2 μm) and macroporous (1–4 μm) perfusion membranes; (b) bleed rate in vessel volume per day, permeate volumetric productivity, and high molecular weight (HMW) species in the bioreactor

4 | CONCLUSIONS

A TFF-based perfusion system, addressing product sieving for a continuous antibody production process, integrated with the capture chromatography step was shown. The macroporous perfusion membrane demonstrated that product sieving coefficient decay can be overcome solely by membrane characteristics in unidirectional TFF perfusion. Nonetheless, other improvements to the operation of the perfusion cell culture bioreactor may also improve filtration performance, such as the addition of shear protectant excipients or more shear resistant cellular hosts. Although the macroporous membrane improved the antibody sieving coefficient, it comes at an expense of increased permeate turbidity, which would require the use of a depth filtration unit operation between the bioreactor and the capture chromatography step. This comprehensive solution, for the interface between the bioreactor and the capture chromatography, yielded a robust aseptic process that minimized filter changeouts.

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REFERENCES

- Belfort, G., Davis, R. H., & Zydney, A. L. (1994). The behavior of suspensions and macromolecular solutions in crossflow microfiltration. *Journal of Membrane Science*, 96, 1–58.
- Blaschczok, K., Kaiser, S. C., Löffelholz, C., Imseng, N., Burkart, J., Bösch, P., ... Eibl, D. (2013). Investigations on mechanical stress caused to CHO suspension cells by standard and single-use pumps. *Chemie Ingenieur Technik*, 85, 144–152.
- Brennan, A. J., Shevitz, J., & Macmillan, J. D. (1987). A perfusion system for antibody production by shear-sensitive hybridoma cells in a stirred reactor. *Biotechnology Techniques*, 1, 169–174.
- Bolton, G. R., & Apostolidis, A. J. (2017). Mechanistic modeling of the loss of protein sieving due to internal and external fouling of microfilters. *Biotechnology Progress*, 33, 1323–1333.
- Clincke, M.-F., Mölleryd, C., Samani, P. K., Lindskog, E., Fäldt, E., Walsh, K., & Chotteau, V. (2013). Very high density of Chinese hamster ovary cells in perfusion by alternating tangential flow or tangential flow filtration in WAVE bioreactor – Part II: Applications for antibody production and cryopreservation. *Biotechnology Progress*, 29, 768–777.
- Clincke, M.-F., Mölleryd, C., Zhang, Y., Lindskog, E., Walsh, K., & Chotteau, V. (2013). Very high density of CHO cells in perfusion by ATF or TFF in WAVE bioreactor. Part I. Effect of the cell density on the process. *Biotechnology Progress*, 29, 754–767.
- de la Broise, D., Noiseux, M., Lemieux, R., & Massie, B. (1991). Long-term perfusion culture of hybridoma: A "grow or die" cell cycle system. *Biotechnology and Bioengineering*, 38, 781–787.
- de la Broise, D., Noiseux, M., Massie, B., & Lemieux, R. (1992). Hybridoma perfusion systems: A comparison study. *Biotechnology and Bioengineering*, 40, 25–32.
- Karst, D. J., Serra, E., Villiger, T. K., Soos, M., & Morbidelli, M. (2016). Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes. *Biochemical Engineering Journal*, 110, 17–26.
- Kim, M.-m., & Zydney, A. L. (2006). Theoretical analysis of particle trajectories and sieving in a two-dimensional cross-flow filtration system. *Journal of Membrane Science*, 281, 666–675.
- Kim, S.-C., An, S., Kim, H. -K., Park, B. -S., Na, K. -H., & Kim, B. -G. (2016). Effect of transmembrane pressure on factor VIII yield in ATF perfusion culture for the production of recombinant human factor VIII co-expressed with von Willebrand factor. *Cytotech*, 68, 1687–1696.
- Klut, S., Magnus, J., Lobedann, M., Schwan, P., Mäyser, B., Niklas, J., ... Schembecker, G. (2015). Developing the biofacility of the future based on continuous processing and single-use technology. *Journal of Biotechnology*, 213, 120–130.
- Konstantinov, K. B., Goudar, C., Ng, M., Meneses, R., Thrift, J., Chuppa, S., ... Naveh, D. (2006). The "Push-to-Low" approach for optimization of high-density perfusion cultures of animal cells. *Advances in Biochemical Engineering/Biotechnology*, 101, 75–98.
- Mariorella, B., Dorin, G., Carion, A., & Harano, D. (1991). Crossflow microfiltration of animal cells. *Biotechnology and Bioengineering*, 37, 121–126.
- Mercille, S., Johnson, M., Lemieux, R., & Massie, B. (1994). Filtration-based perfusion of hybridoma cultures in protein-free medium: Reduction of membrane fouling by medium supplementation with DNase I. *Biotechnology and Bioengineering*, 43, 833–846.
- Patel, B. A., Pinto, N. D. S., Gospodarek, A., Kilgore, B., Goswami, K., Napoli, W. N., ... Richardson, D. D. (2017). On-line ion exchange liquid chromatography as a process analytical technology for monoclonal antibody characterization in continuous bioprocessing. *Analytical Chemistry*, 89, 11357–11365.
- Rathore, A. S., Agarwal, H., Sharma, A. K., Pathak, M., & Muthukumar, S. (2015). Continuous processing for production of biopharmaceuticals. *Preparative Biochemistry and Biotechnology*, 45, 836–849.
- Roush, D. J., & Lu, Y. (2008). Advances in primary recovery: Centrifugation and membrane technology. *Biotechnology Progress*, 24, 488–495.
- Somasundaram, B., Pleitt, K., Shave, E., Baker, K., & Lua, L. H. L. (2018). Progression of continuous downstream processing of monoclonal antibodies: Current trends and challenges. *Biotechnology and Bioengineering*, 115, 2893–2907.
- van Reis, R., Leonard, L. C., Hsu, C. C., & Builder, S. E. (1991). Industrial scale harvest of proteins from mammalian cell culture by tangential flow filtration. *Biotechnology and Bioengineering*, 38, 413–422.
- van Reis, R., & Zydney, A. L. (2007). Bioprocess membrane technology. *J Membrane Sci*, 297, 16–50.
- Vickroy, B., Lorenz, K., & Kelly, W. (2007). Modeling shear damage to suspended CHO cells during cross-flow filtration. *Biotechnology Progress*, 23, 194–199.
- Wang, S., Godfrey, S., Ravikrishnan, J., Lin, H., Vogel, J., & Coffman, J. (2017). Shear contributions to cell culture performance and product recovery in ATF and TFF perfusion systems. *Journal of Biotechnology*, 246, 52–60.
- Warikoo, V., Godawat, R., Brower, K., Jain, S., Cummings, D., Simons, E., ... Konstantinov, K. (2012). Integrated continuous production of recombinant therapeutic proteins. *Biotechnology and Bioengineering*, 109, 3018–3019.
- Xu, S., Jiang, R., Chen, Y., Wang, F., & Chen, H. (2017). Impact of pluronic F68 on hollow filter-based perfusion culture performance. *Bioprocess and Biosystems Engineering*, 40, 1317–1326.
- Yang, W., Minkler, D., Kshirsagar, R., Ryll, T., & Huang, Y. -M. (2016). Concentrated fed-batch cell culture increases manufacturing capacity without additional volumetric capacity. *Journal of Biotechnology*, 217, 1–11.
- Zhang, S., Handa-corrigan, A., & Spier, R. E. (1993). A comparison of oxygenation methods for high-density perfusion cultures of animal-cells. *Biotechnology and Bioengineering*, 41, 685–692.
- Zydney, A. L. (2016). Continuous downstream processing for high value biological products: A review. *Biotechnology and Bioengineering*, 113, 465–475.

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